Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		Because {decay / dice throwing} is random / this smooths out fluctuations in the data / reduces effect of anomalies			1	1		1
		(ii)		1 in 8 should decay **or** $\frac{1}{8}$ of 500 = [62.5 on each throw] (1) So about 60 should be removed **or** 500 – 62.5 = 437.5 (1)			2	2	2	2
	(b)	(i)		All 5 points correctly plotted < 1 small square tolerance (2) 4 points correctly plotted < 1 small square tolerance (1) 3 or less points correctly plotted < 1 small square tolerance (0) Smooth curve of best fit between 0 – 8 throws (1)		3		3	3	3
		(ii)		**At least 1** construction line shown on graph (1) Value correct from candidate's graph – expect 5.1 [throws] (1) If answer is exactly 5.0 accept 5		2		2		2
		(iii)		Fewer dice will decay [on each throw] / 1 in 10 decay [on each throw] / only 50 decay [on the 1st throw] / lower chance of decay [on each throw] (1) Accept converse argument so it will take more throws to remove ½ / so the half-life is longer (1) [so disagree]			2	2		2
				Question 2 total	**0**	**5**	**5**	**10**	**5**	**10**